Day 1: Four parallel assignments

Shared goal for today: Establish whether we can collect trustworthy access data and whether a warm read actually improves a later CUFS read. Those two findings determine what the team can implement and prove.

Engineer	Ownership	Day 1 tasks	Handoff
1 — Protocol / filesystem	CUFS data path and warming	Trace a deployed foreground read through Chimera and CUFS. Identify where file identity, byte range, successful-read outcome, and cold-chunk faulting are available. Test whether a bounded NFS read materializes cold data for a later read.	Confirmed read-path diagram; proposed event-hook location; warm-path findings and gaps.
2 — ML / AI	Dataset contract and prediction	Define the prediction unit and labels with Engineer 1. Design time-of-week and co-access features. Implement an offline evaluator and simple frequency/MRU baselines against fixture data.	Versioned event schema, sample events, label definition, and baseline evaluator.
3 — Data / telemetry / visualization	Collection and data quality	Check existing structured logs and metrics for the schema fields. Identify an approved historical trace. Prototype ingestion, anonymization, loss counters, and a data-quality dashboard using fixtures where necessary.	Field-by-field availability report; trace availability; ingestion/export and quality-check prototype.
4 — Performance	Experiment and proof	Identify a repeatable cold-state reset or equivalent fixture. Build a benchmark skeleton that measures cold read, bounded warm read, and subsequent foreground read. Record latency, faults/remote bytes, and foreground impact where observable.	Reproducible feasibility experiment, raw measurements, and list of missing observability.